

Tenant Requests

Generated Tokens

FairCache

Cache-Aware Fair Scheduler (§IV-B)

Tenant	Waiting-Reqs	CPI-Score
Tenant 1	$R_0 R_1 R_2$	✓
Tenant 2	$R_0 R_1 R_2 R_3$	
Tenant 3		

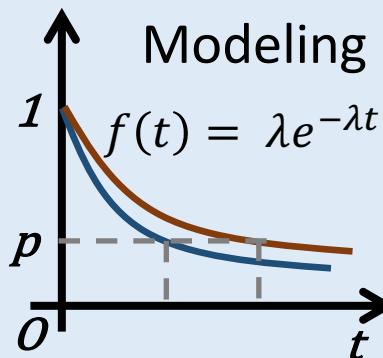
Hit↑ → Cost↓

Freq↑ → Retain↑

Adaptive Fair Cache Manager

Probabilistic Cache Valuation (§IV-D)

Fair Cache Allocation (§IV-C)



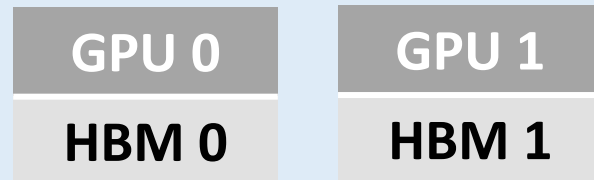
R_0

selected request

allocate
evict

GPU Cluster

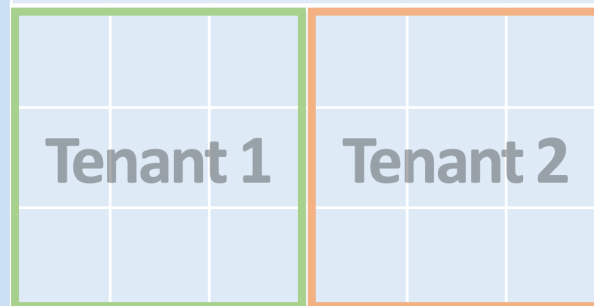
Inference Engine



Load

Store

Host Memory



KV Cache Store

